4003-775 Data Mining
Quarter-Long Team Assignment
Several Due Dates – See Below

Rev. 5/30/2013: Original handout listed project weight as 25% - it should have read 35% (see syllabus)

Overview

For the term project, you will work in teams of 3 to complete a data mining project. You will follow a (nearly) full data mining cycle, including the data selection, cleaning, mining, and analysis.

Using the CRISP-DM as a *guide*, you will:

- Write a problem statement and hypothesis
- Develop a plan for what you want to achieve
- Prepare the data for your processing
- Determine the data mining method(s)
- Build models and analyse the data
- Write a comprehensive report detailing your findings
- Present your findings

Cautions

Get started on this now! It may appear that you don't have enough background to do much, but the initial phases of the project will likely be the most time consuming and tedious.

Be sure you scale your project to something you can complete this 5-week quarter.

Data mining means big data, but you may use a smaller datasets for your first project. You and your team members will locate your data based on what is available and what interests you. I will provide you with some good sources in class.

Carefully read the section **Data Sets** below.

Due Dates

The project represents 35% of your overall grade for the course. Project grading is broken down as follows:

Deliverable	Due Date	Percentage	Who Submits
Team members list	6-2-2013		Team - DB
Team meeting with me to discuss **data**, hypothesis and strategy. ***You must obtain project approval!***	6-6-2013 (Thursday) Make an appointment to see me right after class on this day	Fail or OK to proceed	Team
One page proposal	6-9-2013	5%	Team - DB
Final Report	6-27-2013	80%	Team - DB
Team Presentation	6-25 and 6-27 in class	15%	Team - DB
Team and Self Evaluation	6-27-2013	10 point penalty from your individual report grade if not submitted	Individual - DB

DB = Submit to myCourses Dropbox

Deliverables

** *Be sure all deliverables include the full names of all team members.* I have set up myCourses dropboxes for the different submissions. Be aware that all work will be submitted to a plagiarism checker.

The proposal and progress reports will be evaluated based on the level of detail provided. This doesn't mean they need to be lengthy; I'm more concerned with what you say. *Do not include excessive screen shots of data, graphs, etc. Some of these are OK, but only if you support each graphic with explanations of what it means.*

All written deliverables will be evaluated based on your writing skills, including spelling and grammar. Allow time for you to carefully edit your work!

Details of each deliverable follow:

The **Team Member List** is simply a list consisting of full names and RIT email addresses of the 2-3 people on the team.

The specifics and format of the **Team Meeting** will be discussed in class. You must not proceed until I have approved your data and your problem statement.

The one page **Proposal** must include:

- the full names and RIT email addresses of the team members
- an overview of your project including the hypothesis and goal
- a tentative schedule
- a description of the data file you selected (including a URL to the data)

Your **Final Report** must include:

1. the full names and RIT email addresses of the team members
2. an overview of your project
3. why you felt the problem was important; state your hypothesis
4. how you prepared the data
5. the data mining method(s) you used
6. a comprehensive analysis of your results. Were you able to prove your hypothesis?
7. lessons learned and a discussion of any problems encountered
8. a full and complete References list

I have numbered the items above – **you must use this same numbering to identify each section of your report.** This report represents a large portion of your grade. Do not be disappointed! Be sure you include the 8 numbered sections identified above, and do a thorough job when describing each one. Presentation counts so carefully edit your paper – I will make severe deductions if you submit sloppy work!

Other notes about the final report:

- It is not necessary for you to list all your attributes. It's OK to mention those relevant to your analysis.
- Do not use 0R or 1R for any of your analysis! They are too general and simplistic.
- Add page numbers to your report. I prefer lower right corner.
- Absolutely NO COPY/PASTE without an explanation. I will make drastic deductions if you paste screen shots, confusion matrices, statistics, etc… into your report without a thorough explanation and analysis of what they represent.
- Don't leave it to me to interpret your results – **you tell me** what the results mean.
- Include a printable link to the source of your data (include the actual link in your report; do not embed as a URL).
- Do not make statements such as "This project showed that it always rains on Thursday." Remember you only analysed a very small sampling of data. It's OK to say "This data suggests that it frequently rains on Thursday." This is one example; see me if you don't follow the idea here.

Presentations will be evaluated based on the quality of the content, your ability to describe your project and explain your results, and your team presentation skills. All team members must equally participate in the presentation.

Evaluations - I will provide you with a form you will use you evaluate your own contributions, and the contributions of your team members.

Data Sets

Your team will select the data set. This must be a real, existing data set (nothing made up). To keep things manageable, you are not required to use a realistically sized data set – but your data must consist of at least 7000 instances. Exceptions must be approved by me.

To get a sampling of projects across all the datasets, I will limit the number of teams selecting each one – first come, first assigned.

To ensure that your data will be sufficient, *your team must email me the URL to your data set with a brief description of what you plan to accomplish. I will provide you with feedback – you should not start work until I have approved your data set.*

Some data sets are forbidden:

> You may not use any data involving the US census (the "adult" data file), the Iris database, abalone or mushrooms. In addition, any data set advertising itself as "clean" is prohibited.
>
> If you select any of the IMDB databases, be sure you carefully look at the structure of the sets. Many students have tried and failed to successfully use the IMDB data. Often times, to make meaningful use of the data you need to merge more than one set which is extremely time consuming.
>
> Do not select a dataset with too few attributes.

Important Note

If you find that your team is not working out, let me know ASAP. There is little I can do it you wait until late in the quarter to inform me of a problem. As a courtesy to your team members, let them (and me) know if you plan to withdraw from the course – Please be professional and don't just disappear!

About Plagiarism

Be warned that I will submit all reports to a plagiarism checker. I expect to see in-text citations where warranted, and a full Reference list. I will allow any citation format as long as you are consistent throughout the paper. Any team caught plagiarizing another person’s work will automatically be awarded an F in the course. No exceptions. No excuses. If you aren’t sure about what is allowed, see me.

Rev. 5/17/2013 by tmh